

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B. Tech

Branch : Electronics & Telecommunication Engineering and Allied

Subject Code & Name: Digital Communication/ BTETC602

Semester : VI

Max Marks: 60

Date: 20/12/2024

Duration: 3 Hr

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Objective type questions. (Compulsory Question)

12

- | | | |
|--|-----|---|
| 1 What is necessary for digital communication? | CO1 | 1 |
| a. Precision timing b. Frame synchronization c. Character synchronization d. All of the mentioned | | |
| 2 What are the disadvantages of digital communication? | CO3 | 1 |
| a. Needs more bandwidth b. Is more complex c. Needs more bandwidth & Is more complex d. None of the mentioned | | |
| 3 Synchronization available in digital communication are | CO3 | 1 |
| a. Symbol synchronization b. Frame synchronization c. Carrier synchronization d. All of the mentioned | | |
| 4 Analog to digital conversion includes | CO3 | 1 |
| a. Sampling b. Quantization c. Sampling & Quantization d. None of the mentioned | | |
| 5 The method in which the tail of one pulse smears into adjacent symbol interval is called as | CO2 | 1 |
| a. Intersymbol interference b. Interbit interference c. Interchannel interference d. None of the mentioned | | |
| 6 Random process is a function of _____ | CO1 | 1 |
| a. Random event and time b. Random event and frequency c. Random event and real number d. None of the mentioned | | |

7	Matched filter technique is used to			1
	a. Increase SNR	b. Decrease SNR	c. SNR is not affected	d. None of the mentioned
8	What is the advantage of using matched filter detection?	CO1	1	
	a. Rapid increase in processing gain	b. Simultaneous detection	c. High accuracy	d. Low power consumption
9	FSK reception uses	CO2	1	
	a. Correlation receiver	b. PLL	c. Correlation receiver & PLL	d. None of the mentioned
10	QAM uses _____ as the dimensions.	CO1	1	
	a. In phase	b. Quadrature	c. In phase & Quadrature	d. None of the mentioned
11	The properties used for pseudorandom sequence are	CO4	1	
	a. Balance	b. Run	c. Correlation	d. All of the mentioned
12	The sequences produced by shift register depends on	CO4	1	
	a. Number of stages	b. Feedback tap connections	c. Initial conditions	d. All of the mentioned
Q.2 Solve the following.				12
A)	With a neat block diagram explain the Adaptive Delta Modulation transmitter and receiver with waveforms.	CO3	6	
B)	An analog signal is sampled, quantized and encoded into a binary PCM wave. The sampling frequency is $f_s=8\text{kHz}$ and number of representation levels are 64. The PCM signal is transmitted over a baseband channel using discrete pulse amplitude modulation. Determine the minimum bandwidth of PCM system if each pulse takes, a) 2 amplitude levels, b) 4 amplitude levels or (c) 8 amplitude levels.	CO3	6	
Q.3 Solve the following.				12
A)	Draw the following line code formats for bit stream: 11001001	CO2	6	
	i. Unipolar RZ and NRZ			
	ii. Polar RZ and NRZ			
	iii. Bipolar NRZ (AMI)			
	iv. Split phase Manchester			
	v. Polar Quaternary NRZ			
B)	Derive an expression for maximum signal to quantization noise ratio for linear quantization.	CO1	6	

Q. 4	Solve Any Two of the following.		12
A)	Write a note on Wide-sense stationary process.	CO1	6
B)	Give the classification of Random processes. Explain in detail.	CO1	6
C)	Prove the following autocorrelation properties. i. $R(t)$ is an even function of τ . ii. $R(t)$ is maximum at $\tau=0$.	CO1/CO2	6
Q. 5	Solve Any Two of the following.		12
A)	Describe the generation and detection of coherent QPSK.	CO2	6
B)	Compare all the digital modulation techniques.	CO3	6
C)	Describe the concept of continuous AWGN channel into a vector channel.	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	Define PN sequence. Explain the generation of maximum length sequences along with its properties.	CO3/ CO4	6
B)	Write a note on Personal communication system.	CO1/CO2	6
C)	Write a note on wireless telephone system.	CO1/CO2	6

*** End ***